Table 47. Output voltage characteristics (all I/Os except PC14 and PC15)

The I/O structure options listed in this table can be a concatenation of options including the option explicitly listed. For instance, TT_a refers to any TT I/O with _a option. TT_xx refers to any TT I/O and FT_xx refers to any FT I/O.
All I/Os are CMOS- and TTL-compliant (no software configuration required). Their characteristics cover more than the strict CMOS-technology or TTL parameters. The coverage of these requirements is shown in the FOLLOWING figure.
The minimum and maximum values are specified for a junction temperature (T_J) of 125°C.

Symbol	Parameter	Conditions[2]	Min	Max	Unit
V_{OL}	Output low-level voltage	CMOS port[1], $\lvert I_{IO}\rvert$ = 8 mA, 2.7 V ≤ V_{DD} ≤ 3.6 V	-	0.4	V
V_{OH}	Output high-level voltage	CMOS port[1], $\lvert I_{IO}\rvert$ = -8 mA, 2.7 V ≤ V_{DD} ≤ 3.6 V	V_{DD} - 0.4	-	
V_{OL}[2]	Output low-level voltage	TTL port[1], $\lvert I_{IO}\rvert$ = 8 mA, 2.7 V ≤ V_{DD} ≤ 3.6 V	-	0.4	
V_{OH}[2]	Output high-level voltage	TTL port[1], $\lvert I_{IO}\rvert$ = -8 mA, 2.7 V ≤ V_{DD} ≤ 3.6 V	2.4	-	
V_{OL}[2]	Output low-level voltage	$\lvert I_{IO}\rvert$ = 20 mA, 2.7 V ≤ V_{DD}≤ 3.6 V	-	1.3	
V_{OH}[2]	Output high-level voltage	$\lvert I_{IO}\rvert$ = -20 mA, 2.7 V ≤ V_{DD}≤ 3.6 V	V_{DD} - 1.3	-	
V_{OLFM+}[2]	Output low-level voltage for a FT_f I/O pin in FM+ mode	$\lvert I_{IO}\rvert$ = 20 mA, 2.7 V ≤ V_{DD} ≤ 3.6 V	-	0.4	

1. *TTL and CMOS outputs are compatible with JEDEC standards JESD36 and JESD52.*
2. *Specified by design - Not tested in production.*

Table 48. Output voltage characteristics for PC14 and PC15

Specified by design and not tested in production.
The minimum and maximum values are specified for a junction temperature T_J = 125°C.
The IIO current sourced or sunk by the device must always comply with the absolute maximum rating specified in Table 14. Current characteristics. Additionally, the sum of the currents sourced or sunk by all the I/Os (I/O ports and control pins) must always comply with the absolute maximum ratings ΣIIO.

Symbol	Parameter	Conditions(3)	Min	Max	Unit
VOL	Output low level voltage	CMOS port[1] IIO=0.5 mA 2.7 V≤ V_{DD} ≤3.6 V	-	0.4	V
VOH	Output high level voltage	CMOS port[1] IIO=-0.5 mA 2.7 V≤ V_{DD} ≤3.6 V	V_{DD} - 0.4	-	
VOL[2]	Output low level voltage	TTL port[1] IIO = 0.5 mA 2.7 V ≤ V_{DD} ≤3.6 V	-	0.4	
VOH[2]	Output high level voltage	TTL port[1] IIO = -0.5 mA 2.7 V ≤ V_{DD} ≤ 3.6 V	2.4	-	

1. *TTL and CMOS outputs are compatible with JEDEC standards JESD36 and JESD52.*
2. *Specified by design - Not tested in production.*

Output AC characteristics

The definition and values of output AC characteristics are given in Figure 21. Output AC characteristics definition and in the table below respectively.

Unless otherwise specified, the parameters given are derived from tests performed under the ambient temperature and supply voltage conditions summarized in Table 16.